

Results

Weights dynamics

We look at the normalized eigenvectors of $E(t) = \begin{pmatrix} \mu(t) & -\beta\nu(t) \\ -\nu(t) & \beta\mu(t) \end{pmatrix}$

$$\vec{V}_+(t) = \frac{\mu(t)}{N_+(t, \beta)} \begin{pmatrix} 1 \\ \frac{\nu(t)}{\beta\mu(t) - \lambda_+(t, \beta)} \end{pmatrix} = \frac{1}{N_+(t, \beta)} \begin{pmatrix} V_1^+(t) \\ V_2^+(t) \end{pmatrix}$$

$$\vec{V}_-(t) = \frac{\mu(t)}{N_-(t, \beta)} \begin{pmatrix} \frac{\beta\nu(t)}{\mu(t) - \lambda_-(t, \beta)} \\ 1 \end{pmatrix} = \frac{1}{N_-(t, \beta)} \begin{pmatrix} V_1^-(t) \\ V_2^-(t) \end{pmatrix}$$

With eigenvalues:

$$\lambda_{\pm}(\beta, t) = \frac{\mu(t)(1 + \beta) \pm \sqrt{\mu^2(t)(1 - \beta)^2 + 4\beta\nu^2(t)}}{2}$$

where:

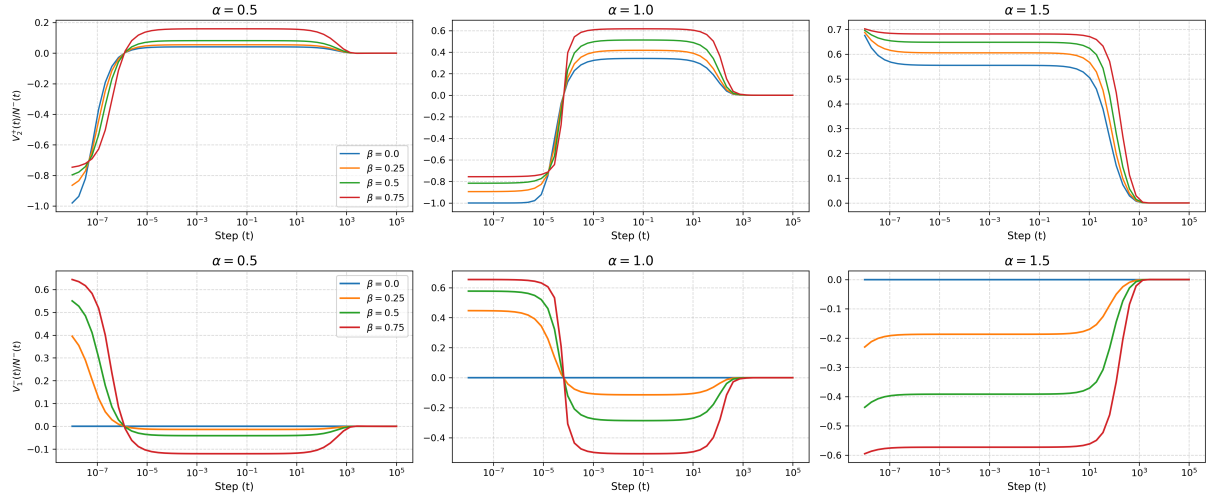
$$\mu(t) = 1 - \sum_{i=1}^d \pi_i e^{-\eta \pi_i t} \quad \nu(t) = \sum_{i=1}^d \pi_i u_i e^{-\eta \pi_i t}$$

Normalized off-diagonal components dynamics for different vocabulary sizes d and α .

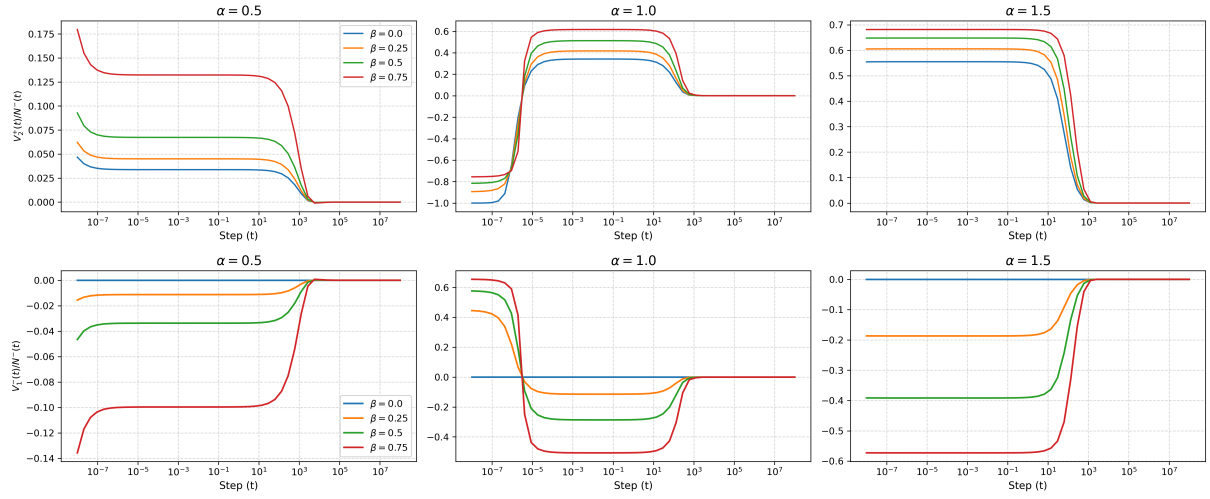
$$V_2^+(t) = \frac{\mu(t)\nu(t)}{N_+(t)(\beta\mu(t) - \lambda_+(t))} \quad V_1^-(t) = \frac{\beta\mu(t)\nu(t)}{N_-(t)(\mu(t) - \lambda_-(t))}$$

Greedy sampling:

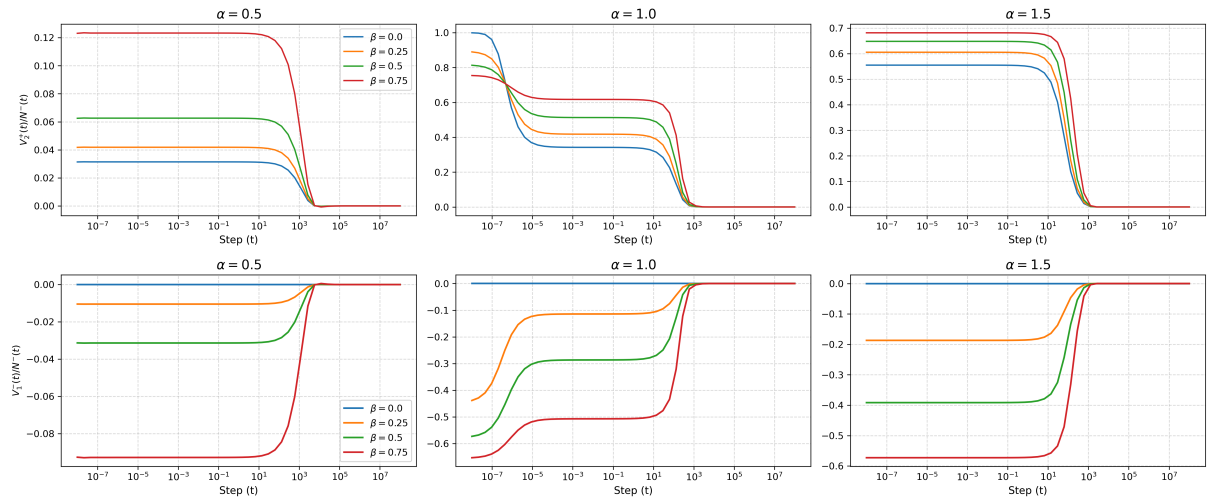
Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (greedy sampling, $d = 1000$)



Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (greedy sampling, $d = 5000$)

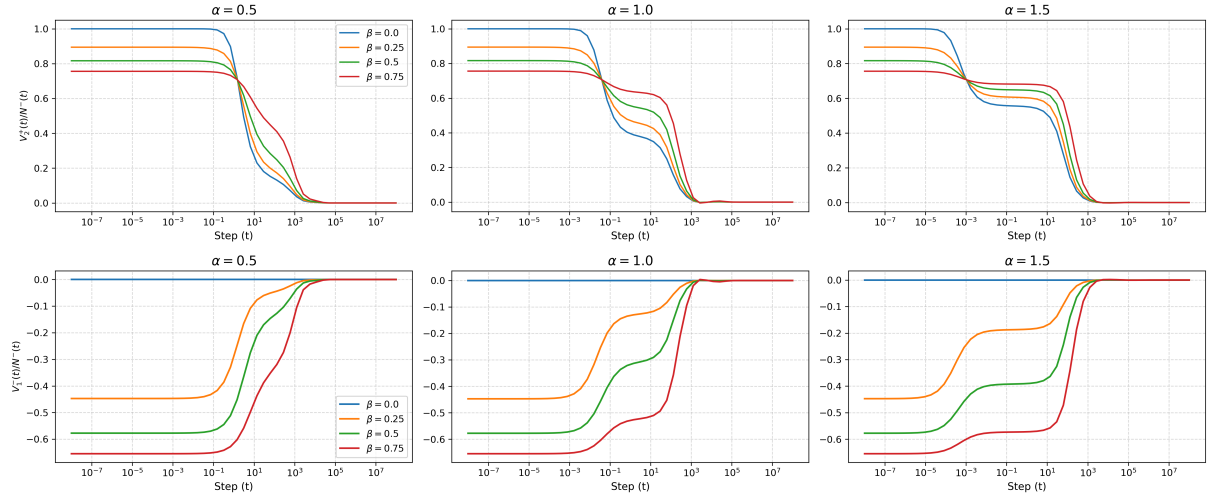


Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (greedy sampling, $d = 10000$)

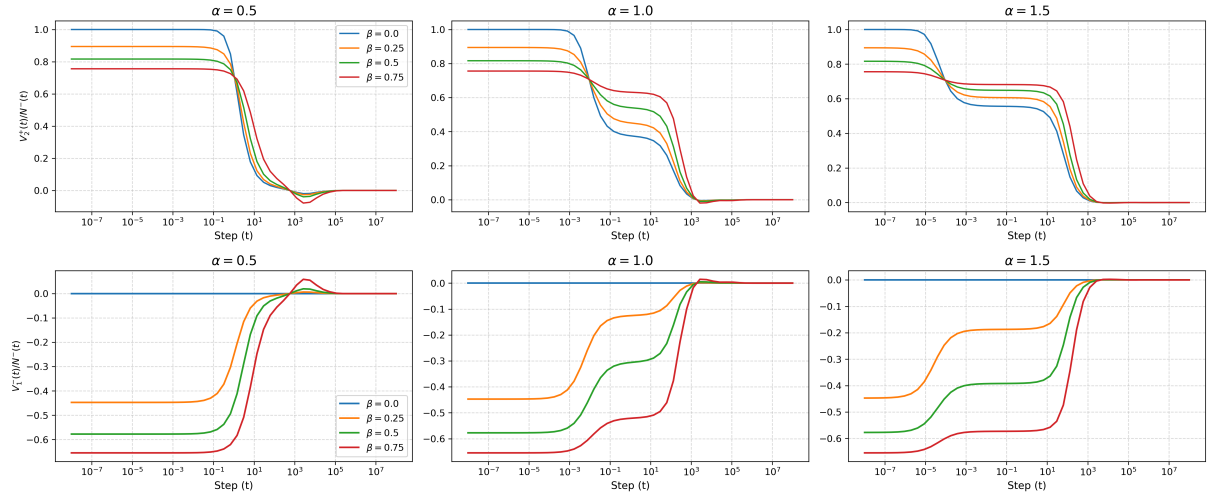


Flip sampling (10 passes)

Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (flip sampling, $d = 1000$)



Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (flip sampling, $d = 5000$)



Dynamics of normalized off-diagonal components $V_2^+(t)$ and $V_1^-(t)$ (flip sampling, $d = 10000$)

